Topic D: written papers and complete programs

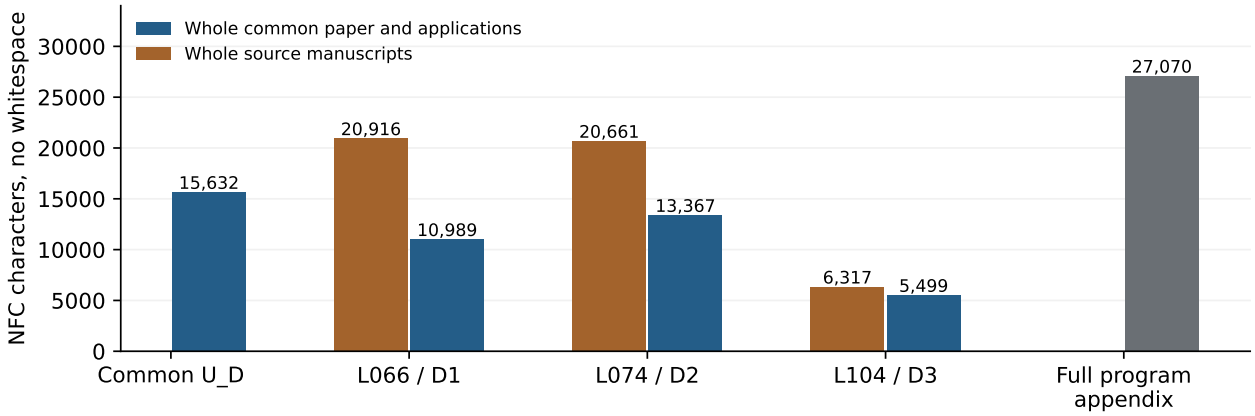
Human Compression project

The three accepted applications recover the declared principal results on commuting involutions (L066/D1), Suzuki regular polyhedra (L074/D2), and Hall–Janko geometries (L104/D3). The common paper supplies orbit data, stabilizer fibers, support counting and panel recognition. The exact 37-row source contract, corrections, retained inputs and exclusions remain unchanged. This is a comparison of complete extracted manuscript text for those main results, not equivalence of every printed assertion.

Written papers. The three whole source manuscripts cost 47,894 NFC characters without whitespace. Our whole common paper costs 15,632 and the three whole applications cost 29,855. Charging the common paper once within Topic D gives

$$R_D = \frac{\sum_i c(L_i)}{c(U_D) + \sum_i c(D_i)} = \frac{47\,894}{15\,632 + 29\,855} = \frac{47\,894}{45\,487} = \mathbf{1.053}.$$

The common paper is paid in full even where another topic uses the same idea.



All bars use complete documents after exact running-furniture removal and occurrence-specific visible-glyph calibration. Unique front matter and references remain. The grey program bar is a separate comparison below. These document costs are not a time series or evidence of a sawtooth pattern.

After acquiring the common work. After acquiring *the whole common paper and the complete program boundary*, the conditional comparison is $c(L_i)/c(D_i)$. That boundary includes all five required programs and all three complete diagnostics. Both common theory and all eight programs are acquired in full before these local comparisons.

Source / application	Whole source	Whole application	Conditional factor
L066 / D1	20,916	10,989	1.903
L074 / D2	20,661	13,367	1.546
L104 / D3	6,317	5,499	1.149

These conditional factors do not include the acquisition cost a second time. The combined written comparison above pays the whole common paper explicitly; the next comparison also pays every printed program.

Including the required program. The complete 12-page appendix contains all eight mathematical files: five required programs and three full diagnostics. It costs 27,070 in the primary unit, so the separate known-text comparison is

$$R_{D,\text{program}} = \frac{47\,894}{45\,487 + 27\,070} = \frac{47\,894}{72\,557} = \mathbf{0.660}.$$

Here the replacement is longer than the available historical manuscript text. Historical software and external algorithm construction have unknown cost; they are not assumed to be free. The comparison states exactly which measured texts are in its numerator and denominator.

Representation	Denominator	Factor
Written papers	45,487	1.053
Written papers plus whole printed appendix	72,557	0.660
Written papers plus all eight whole raw programs	71,672	0.668

The whole raw programs cost 26,185, of which 17,209 belongs to the five required files and 8,976 to the three diagnostics. This is an *alternative* to the printed appendix, never an additional charge. Raw metrics are summed file by file, without pruning branches or diagnostic parts. The two complete execution drivers cost 10,038 separately and are reported separately from the denominators above. Their 12,765 actual file bytes are also retained in the machine-readable inventory. One driver retains an unused historical M22 branch; no M22 result is included.

Sensitivity to the ruler. The full tables retain three variants (raw untrimmed, raw selected and calibrated selected), NFC and NFKC, and five units: 30 coordinates and 180 exact combined/conditional fractions. The middle columns below use calibrated selected text; the range covers all six variants/normalizations.

Unit	Written NFC	Written NFKC	Written range	With code NFC
Whitespace tokens	1.150	1.153	1.150–1.160	0.895
Alphanumeric runs	1.237	1.237	1.237–1.245	0.802
Nonspace characters	1.053	1.056	1.053–1.064	0.660
Collapsed characters	1.069	1.072	1.069–1.080	0.693
Collapsed UTF-8 bytes	1.020	1.091	1.016–1.099	0.680

No one of these units measures intrinsic mathematical difficulty. All use the same complete-document and program boundary.

Extent and limits. The three source PDFs have 9, 12 and 5 physical pages (26 total). Our common paper has 6 pages, the applications have 5, 6 and 2 (19 scientific pages total), and the program appendix has 12. All 57 input pages are retained. Vector edges, formula placement and other unextracted graphical information have unknown cost. This is an extracted-text coordinate, not a claim to count every visible mark or all prior research effort. External classical proofs, subgroup catalogue construction, GAP internals and formal verification costs are separate.

The L104 printed diagrams, full Aut/Cor groups, doubling relation and supplement annotations are not all recovered; their source text is nevertheless part of the complete manuscript numerator. The L066 reversal/regularity qualifications and the retained Suzuki classification remain explicit. The corrected extraction uses original PDF bytes for fourteen L066 brackets previously confused with brace pieces. This is an extracted-character/encoding collision, not a ToUnicode-table collision: that font has no such stream. Only calibrated UTF-8 lengths changed by -28; the primary character counts did not change. Input acceptance and numerical admission are recorded separately. No new source, conjecture or formal theorem credit follows from this measurement.